AMD DEVELOPER HACKATHON: ACT II · TRACK 2

CaptionForge

One clip in. Four captions out — each written in its own voice.

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THE CHALLENGE

Watch a clip. Caption it four ways.

Track 2 asks an agent to watch a short video and write one caption per requested style — then a judge scores every caption.

≤ 10 min	~12 clips	2 scores
total runtime for the whole container	hidden set, 30s–2min each	Accuracy (0–1) + Style match (0–1)

Every clip must be captioned in all four styles:

- formal
- sarcastic
- humorous_tech
- humorous_non_tech

A missing style scores zero for the entire clip — so the pipeline never leaves one blank.

THE BIG IDEA

One clean call per style

V6 cuts the relay: instead of a describer, a writer, and a judge passing the clip along, one model sees the frames and writes the caption directly — guarded by deterministic code instead of a second AI opinion.

WRITES

qwen3p7-plus

Sees the sampled frames and the style brief together, replies with the finished caption inside a strict tag.

BACKS UP

kimi-k2p7-code

A different model family, held in reserve — only called if the primary is still down after retries.

GUARDS

plain code

Regex catches emoji, banned slang, and a missing tech term — no LLM judge needed to enforce style rules.

Same benchmark process as before — head-to-head evaluation, cross-judged by neutral models on the official rubric.

HOW IT WORKS

From video URL to four captions

1	**Download**	Fetch the clip with retries and a hard time cap so a slow server can't stall the run.
2	**Sample frames**	Pull 4 evenly-spaced frames from the clip — straight to the vision model, no scene-report stage in between.
3	**Caption, per style**	One multimodal call per style, in parallel — the model replies with the caption inside a strict output tag.
4	**Guard & retry**	A missing tag or a broken style rule triggers one retry, then a regeneration naming the exact problem.
5	**results.json**	Every task, every style filled in — the one file the judge reads.

THE FOUR VOICES

Same clip, four personalities

formal

30–50 words

A wire-service photo desk: neutral, precise, one observational read on the frame — no opinion, no flourish.

sarcastic

15–32 words

Dry, deadpan irony. Mocks something specific in the clip, never hyped-up.

humorous_tech

18–35 words

One real tech term required — maps the scene onto software life, the actual subject stays readable.

humorous_non_tech

18–35 words

The funny uncle at dinner — one warm, everyday comparison. Zero technology words allowed.

ACCURACY GUARDRAILS

Designed not to make things up

The biggest way a vision model loses points is inventing detail. Our prompts and code block that on purpose.

✓ No guessed specifics

Never names a real city, country, or landmark — only the type of place, in plain words.

✓ No over-read text

Never quotes or transcribes on-screen text — a misread sign costs more than a generic description.

✓ Code, not a second opinion

Emoji, banned slang, a missing tech term, a stray '!' in formal — all caught by regex, not an LLM judge.

✓ Fix named, not guessed

A caption that still breaks a rule gets exactly one regeneration, with the specific problem spelled out.

BUILT TO FINISH

It always produces a valid result

The 10-minute clock is a hard rule. CaptionForge is engineered so nothing — a stuck clip, a bad key, a broken input — can stop it from finishing cleanly.

1. **Race the clock** — Clips are sized and processed heaviest-first, against a hard wall-clock deadline.
2. **Recover, then fallback** — A failed call gets a paced retry pass with leftover time budget; only then does a style fall back to a tone-correct default — never a zero.
3. **Can't crash out** — Bad input or a missing key still writes a valid results.json and exits cleanly.

WHY CAPTIONFORGE

Smart where it counts, safe under pressure

One clean call

Each style gets one direct model call instead of a multi-stage relay — less latency, fewer points of failure.

Guarded by code

Deterministic checks enforce every style rule and trigger automatic regeneration — no second AI needed to police the first.

Accuracy by design

Guardrails against invented detail live in both the prompts and the code.

Bulletproof

A failed call gets a paced retry, not a permanent fallback — still finishes inside 10 minutes with a fully-filled result.